

Numerical Study of Multi-Matrix Models

(Project Report)



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Abstract

This report presents the preliminary analysis of Hoppe's matrix model using numerical computations and compares our results with analytical forms given in Ref. [\[1\]](#). We used the Metropolis Monte Carlo algorithm to evolve the fields of the model. Our results in the strong coupling regime match nicely with the analytical results.

Acknowledgements

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1 What is Metropolis Monte Carlo Method?

In nature, we can find true random events such as radioactive decay, atmospheric noise, thermal noise, and roulette wheel. However, when we generate random numbers on a computer using an algorithm, no two random numbers are truly random to each other. The Monte Carlo method is an algorithm that needs repeated random sampling to obtain numerical results.

The Monte Carlo method can use random numbers to solve very deterministic problems. Markov chain Monte Carlo method is a set of algorithms that sample from probability distributions. It can characterize a mathematical distribution without knowing the properties of that distribution just by randomly sampling values from the distribution. The Metropolis Monte Carlo method is one type of Markov chain Monte Carlo method.

The Metropolis method initializes a system with some random values (or all zeroes also). Then a shift is computed and added to the initial system. This process results in a new system that is slightly shifted away from the original system. Once shifted, the energy difference between the new and original systems is compared. If the energy difference is negative, the new system is accepted, and it is accepted with a conditional probability if the energy difference is positive. This acceptance and rejection are done several million times, and finally, the average of the observables is calculated.

In this report, we will take the results given in Ref. [1] and analyze them with the Metropolis Monte Carlo method. First, we will derive all the equations we need.

2 Finding the Observables We Need

Let us consider the two matrix model first, with a commutator interaction. The partition function for the same is given as:

$$Z = \int DXDY e^{-\text{Tr}X^2 - \text{Tr}Y^2 + g^2 \text{Tr}[X,Y]^2}, \quad (1)$$

where X and Y are traceless Hermitian matrices and they interact via a commutator potential term with a coupling parameter g . This integral is quadratic in both X and Y , so we can diagonalize X and perform the integral over Y alone. Once we simplify the partition function we will get logarithmic terms, which can be solved

with the standard Vandermonde determinant that comes from diagonalizing X . A large- N saddle point to this integral can be obtained. It looks like this:

$$x_i = \sum_{j \neq i} \frac{1}{(x_i - x_j)[1 + g^2(x_i - x_j)^2]}, \quad (2)$$

where x_i are the saddle point equations of motion. This has been solved at large N by [2] implicitly. We will be taking the strong t'Hooft coupling limit, i.e., $\lambda = g^2 N \gg 1$. In the large- N limit, the saddle point equation becomes

$$x = \int \frac{\rho(y)dy}{(x - y)[1 + g^2(x - y)^2]}, \quad (3)$$

where $\rho(y)$ is an eigenvalue density function. We can solve this integral using the general results in Ref. [1], which gives us

$$\rho(x) = \frac{3N(L^2 - x^2)}{4L^3}, \quad (4)$$

where

$$L = N^{1/2} \frac{3\pi^{1/3}}{2^{1/3}\lambda^{1/6}}. \quad (5)$$

Now, it is easy to evaluate the observables

$$\frac{\text{Tr} X^2}{N} = \int_{-L}^L \frac{3x^2(L^2 - x^2)}{4L^3} dx = \frac{L^2}{5} = \left(\frac{3\pi}{2}\right)^{2/3} \frac{N}{5\lambda^{1/3}} + \dots. \quad (6)$$

To find out if the two matrices commute at strong coupling, the natural observables to use are commutators. So, along with commutators we are interested in the following set of observables:

$$\text{Tr}(XYXY), \quad \text{Tr}(X^2Y^2), \quad \text{Tr}[X, Y]^2.$$

Using the this set of observables, we define the following ratio, which will tell us about commutativity of the two matrices

$$r = \frac{\text{Tr}(XYXY)}{\text{Tr}(X^2Y^2)}. \quad (7)$$

It will serve as an order parameter that can tell us about the correlation of the matrices. We know that:

$$\text{Tr}[X, Y]^2 = N \frac{\partial \log Z}{\partial \lambda}. \quad (8)$$

Using the expression we obtained for $\rho(x)$ in Eq. 4 we can solve the above equation to obtain

$$\text{Tr}(X^2) = \frac{-N^3}{2\lambda} + \dots. \quad (9)$$

Further, to obtain the value of the other observables, we can introduce a source J_{ij} to the partition function and thus we get

$$Z = \int DX DY e^{-\text{Tr}X^2 - \text{Tr}Y^2 + g^2 \text{Tr}[X, Y]^2 + \text{Tr}JY}. \quad (10)$$

Upon diagonalizing the matrix X and then solving, we can get expressions for $\text{Tr}(XYXY)$ and $\text{Tr}(X^2Y^2)$ in the integral form. Upon solving those integrals, we obtain

$$\text{Tr}(XYXY) = \frac{3\pi}{70} \left(\frac{3\pi}{2} \right)^{1/3} \frac{N^3}{\lambda^{2/3}} - \frac{3N^3}{8\lambda} + \frac{\alpha N^3}{\lambda} + \dots \quad (11)$$

and

$$\text{Tr}(X^2Y^2) = \frac{3\pi}{70} \left(\frac{3\pi}{2} \right)^{1/3} \frac{N^3}{\lambda^{2/3}} - \frac{N^3}{8\lambda} + \frac{\alpha N^3}{\lambda} + \dots. \quad (12)$$

The last term of both the above equations, which contains α , comes from the leading correction of the $\rho(x)$ equation.

2.1 What Do the Observables Tell Us?

The observables for $\text{Tr}(XYXY)$ and $\text{Tr}(X^2Y^2)$ can tell us about the commutativity of the two matrices at strong coupling. Consider the following expressions

$$N \frac{\text{Tr}[X, Y]^2}{\text{Tr}X^2 \text{Tr}Y^2} = -\frac{25}{2} \left(\frac{2}{3\pi} \right)^{4/3} \frac{1}{\lambda^{1/3}} + \dots \rightarrow 0 \quad (13)$$

and

$$\frac{\text{Tr}(XYXY)}{\text{Tr}(X^2Y^2)} = 1 - \frac{35}{6\pi} \left(\frac{2}{3\pi} \right)^{1/3} \frac{1}{\lambda^{1/3}} + \dots \rightarrow 1. \quad (14)$$

The reason we take these two ratios as the observables we need is because if we want to talk about the commuting nature of the matrices X and Y in the strong coupling regime then we have to take observables that have both X and Y in them. The most obvious choice for this are commutators. Further, $\text{Tr}(XYXY)$ would vanish at the planar level whereas $\text{Tr}(X^2Y^2)$ are non-vanishing. The observable $\text{Tr}[X, Y]^2$ will tell us how close to commuting these matrices are. The value of $\lambda^{1/3}$ controls the size of the non-commutativity of matrices. The above two equations are a criteria for the emergence of a local geometry [1].

3 A General Zero-Dimensional Model

The action of a zero-dimensional two-matrix model that we need can be obtained by the expression of the partition function given above.

We have

$$S = N \text{Tr} \left(X^2 + Y^2 - g^2[X, Y]^2 + h^2X^4 + h^2Y^4 \right), \quad (15)$$

where X and Y are $N \times N$ complex Hermitian matrices, and g and h are coupling parameters. When h is 0 and g is finite, the model reduces to that introduced by Hoppe [3] and investigated in detail in Ref. [2]. In this report we will only work with the original Hoppe model. The action when $h = 0$ becomes:

$$S = N \text{Tr} \left(X^2 + Y^2 - g^2[X, Y]^2 \right), \quad (16)$$

where $g = \sqrt{\frac{\lambda}{N}}$. These are the observables we are interested in, and as seen above,

their analytical forms are:

$$\frac{\text{Tr} X^2}{N} = \frac{(3\pi)^{2/3} N}{5 \cdot 2^{2/3} \lambda^{1/3}} + \dots \quad (17)$$

$$\text{Tr}[X, Y]^2 = -\frac{N^3}{2\lambda} + \dots \quad (18)$$

$$N \frac{\text{Tr}[X, Y]^2}{\text{Tr} X^2 \text{Tr} Y^2} = -\frac{25}{2} \left(\frac{2}{3\pi} \right)^{4/3} \frac{1}{\lambda^{1/3}} + \dots \quad (19)$$

$$\frac{\text{Tr}(XYXY)}{\text{Tr}(X^2Y^2)} = 1 - \frac{35}{6\pi} \left(\frac{2}{3\pi} \right)^{1/3} \frac{1}{\lambda^{1/3}} + \dots \quad (20)$$

4 The Metropolis Monte Carlo Algorithm

The Metropolis Monte Carlo (MMC) algorithm can generate configurations that follow the following probability distribution:

$$p(v) = \frac{1}{Z} e^{-\beta f(v)}, \quad (21)$$

where $f(v)$ is a function which is generally the action or the energy of the state, and ν and β are parameters of the model.

The algorithm first has to produce random numbers. As we know, numbers can never be truly random and whenever a computerized program tries to generate random numbers, it does so with a starting seed. Hence, the starting points can be different with different seeds, but eventually our fields should converge to the same value with all different seeds. It means we need to truncate the initial information, which is seed dependent, and only keep the seed independent data, which known as *thermalization*.

4.1 What is Thermalization?

As discussed above, the method that most random functions use for generating random numbers is such that the series generated will start with numbers close to the seed and then after some generations the numbers will start to branch out into other parts of the domain within which the numbers are expected to be generated. So to keep the entire series absolutely random, we have to neglect the first few numbers that have gotten an ‘undue advantage’ whilst being produced. This process is called thermalization. The algorithm for Metropolis Monte Carlo always starts with seed

dependent results and then moves onto thermalized results. After removing initial configurations, we get the thermalized data which is used in the data analysis part.

In this report, we will be analyzing a zero-dimensional model and in the future, we will move to a general one-dimensional model with two Hermitian matrices. We will be using the MMC algorithm to generate our configurations. Also, we will try to shift the matrices such that we get a reasonable acceptance and evolution.

Let us see the acceptance rate and Monte Carlo (MC) evolution for one of the target configuration in our model. In Fig. 1 we show the acceptance rate and $\text{Tr}(X^2)$ against MC time for $N = 5$ and $\lambda^{1/3} = 0.5$.

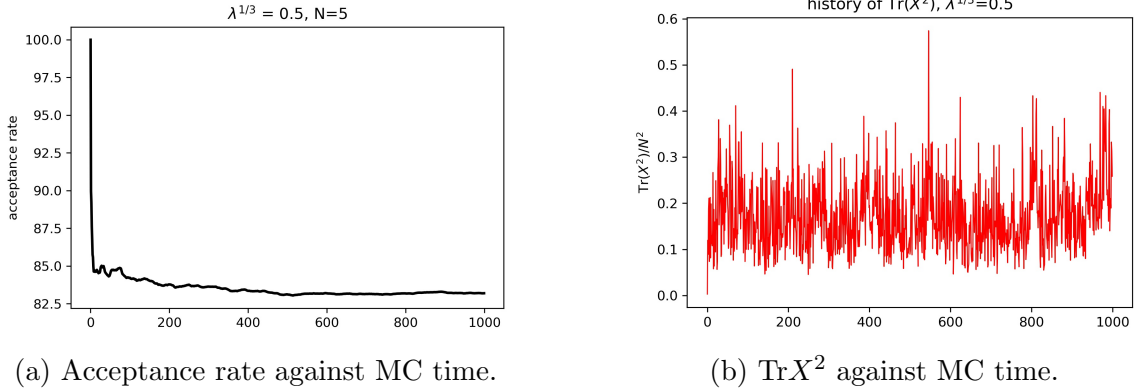


Figure 1: Acceptance rate and $\text{Tr}(X^2)$ against MC time for $N = 5$ and $\lambda^{1/3} = 0.5$.

As can be seen that with $\lambda^{1/3} = 0.5$ we keep the acceptance rate around 80% and hence the Monte Carlo time evolution of one such observable $\text{Tr}X^2$ is reasonable, and we will get thermalized data set by truncating some initial configurations.

5 Results

We first work with $N = 5$ and see how comparable the observed results are with respect to the analytical ones. The results are shown in Tabs. 1 and 2 for various observables. In Tab. 1 we show the computed averaged values of observables defined by Eqs. (17), (18), (19), and (20) for different λ values with $N = 5$, and in Tab. 2 we show the same for $N = 10$.

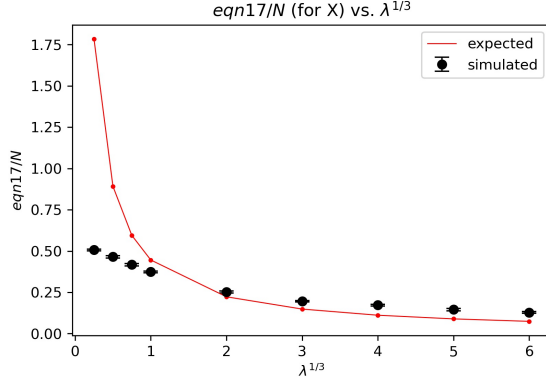
$\lambda^{1/3}$	Eq. (17)/N	Eq. (17)/N (for Y)	Eq. (18)/ N^3	Eq. (19)	Eq. 20
0.25	0.5068(57)	0.4971(66)	-0.4643(92)	-3.68(48)	0.068(14)
0.5	0.4650(70)	0.4618(52)	-0.3440(53)	-3.20(39)	0.172(16)
0.75	0.4172(71)	0.4152(54)	-0.2254(31)	-2.60(33)	0.289(18)
1	0.3742(49)	0.3546(40)	-0.1469(20)	-2.21(24)	0.383(19)
2	0.2523(56)	0.2400(48)	-0.03295(45)	-1.08(18)	0.677(41)
3	0.1953(4))	0.1930(54)	-0.01159(10)	-0.61(11)	0.816(44)
4	0.1726(53)	0.1627(30)	-0.00528(6)	-0.376(71)	0.895(59)
5	0.1455(73)	0.1254(52)	-0.00282(3)	-0.30(10)	0.910(96)
6	0.1278(50)	0.1342(47)	-0.00167(2)	-0.194(53)	0.942(66)

Table 1: Computed averaged values of observables defined by Eqs. (17), (18), (19), and (20) for different λ values with $N = 5$.

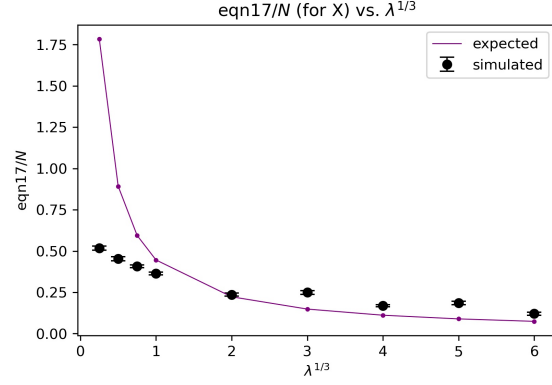
$\lambda^{1/3}$	Eq. (17)/N	Eq. (17)/N (for Y)	Eq. (18)/ N^3	Eq. (19)	Eqn. 20
0.25	0.517(12)	0.523(12)	-0.489(25)	-1.80(11)	0.067(21)
0.5	0.453(12)	0.492(15)	-0.348(12)	-1.55(86)	0.197(28)
0.75	0.407(7)	0.392(8)	-0.2302(44)	-1.44(50)	0.228(26)
1	0.364(7)	0.336(9)	-0.1459(33)	-1.19(50)	0.334(26)
2	0.236(9)	0.248(9)	-0.03307(58)	-0.56(31)	0.663(46)
3	0.249(11)	0.190(9)	-0.01150(14)	-0.24(14)	0.862(91)
4	0.168(6)	0.134(6)	-0.0053(74)	-0.23(14)	0.865(96)
5	0.185(10)	0.139(6)	-0.0027(34)	-0.108(76)	0.94(14)
6	0.1203(86)	0.162(8)	-0.0016(17)	-0.086(77)	0.95(14)

Table 2: Computed averaged values of observables defined by Eqs. (17), (18), (19), (20) for different λ values with $N = 10$.

To compare the analytical values and the simulated values, we have plotted the data for configurations with $N = 5$ and $N = 10$ matrix sizes. The plots are shown in Figs. 2, 3, 4, 5, and 6.

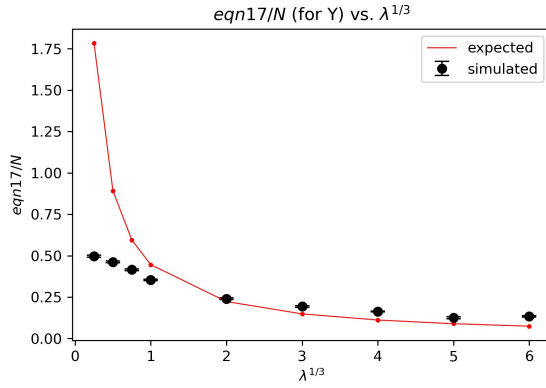


(a) $N = 5$

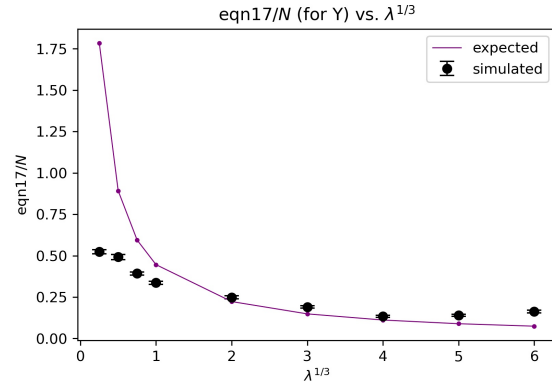


(b) $N = 10$

Figure 2: Eq. (17) / N vs $\lambda^{1/3}$ with different N values for matrix X .

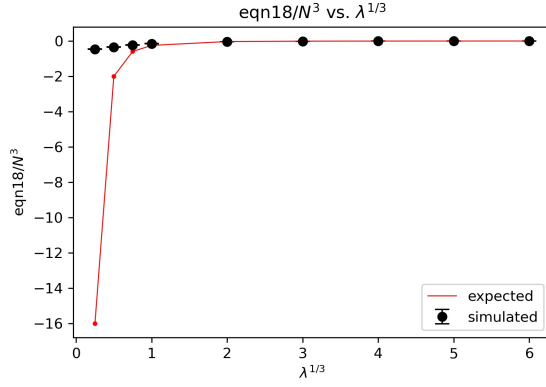


(a) $N = 5$

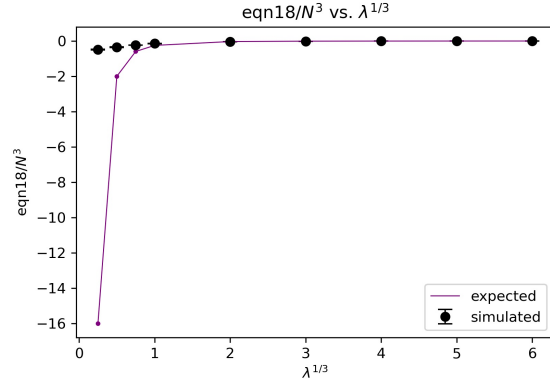


(b) $N = 10$

Figure 3: Eq. (17) / N vs $\lambda^{1/3}$ with different N values for matrix Y .

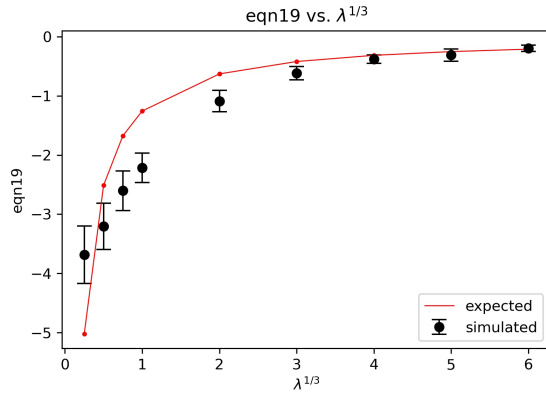


(a) $N = 5$

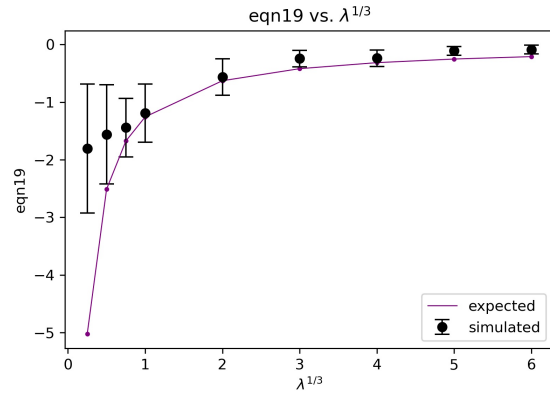


(b) $N = 10$

Figure 4: Eq. (18) / N^3 vs $\lambda^{1/3}$ with different N values.



(a) $N = 5$



(b) $N = 10$

Figure 5: Eq. (19) vs $\lambda^{1/3}$ with different N values.

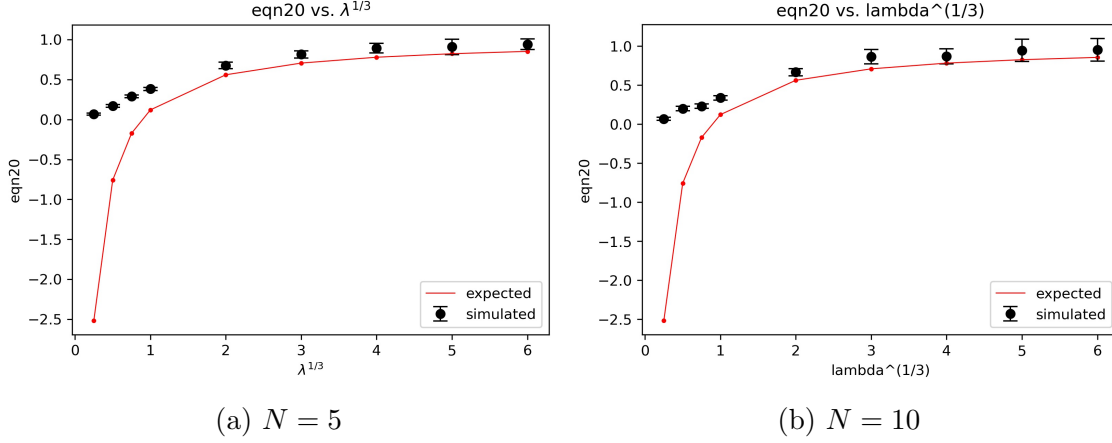


Figure 6: Eq. (20) vs $\lambda^{1/3}$ with different N values.

6 Observations

Since $\lambda = g^2 N$, we change λ whilst keeping N constant, then g will get modified. So the analytical results are expected to match with our observations only in the strong coupling regime as analytical results have truncations at first term, which have some powers of λ in the denominator. Also, in analytical results we rescaled λ to 2λ .¹

So, in the above graphs when we changed λ , the first 3-4 points deviated drastically from the analytical curves as expected. This is because for those points, the simulated points are in the weak coupling regime and simulated results are on the expected lines for the strong coupling regime.

7 Conclusions

We can clearly see that for most of the points, the simulated and the analytical values are very close in the strong coupling regime. This implies that we have been able to confirm the observables given in Ref. [1] with numerical results using Metropolis Monte Carlo method.

¹The reason as to why this works is still being studied. This report is still in the initial stages of the work.

8 Future Directions

To continue this work, we plan to analyze a general one-dimensional model in the same way and obtain numerical results for the same.

We also plan to look into the large- N behavior, and the distribution of the eigenvalues, and comparing those to its analytical distribution for this zero-dimensional model itself.

References

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